



EDUCATION AND SCHOLASTIC ACHIEVEMENTS

Program	Institute	Year
Electrical Engineering (Dual Degree)	Indian Institute of Technology Madras	2021
Class XII (CBSE)	D.A.V Public School, Kota	2016
Class X	St. Marys' School, Pune	2014

- Secured a rank of in IIT-JEE Advance, 2016.
- Awarded **Gold Medal** in NSO (National Science Olympiad) in class X and selected for **2nd stage**.
- Selected for **2nd stage** in IMO (International Mathematics Olympiad) in class X.

COURSEWORK AND SKILLS

- Circuits and Networks
- Analog Systems and Lab
- Digital Systems and Lab
- Analog Circuits
- Programming Languages: **Python and C**
- Softwares: **Cadence Virtuoso, LTspice, Eldo, Electric, Matlab.**
- Digital IC Design
- Analog IC Design
- Phase Locked Loops
- Solid State Devices
- Power Electronics
- Electric Machines and Lab
- Signals and Systems
- Microprocessor Theory
- Probability and Statistics
- Communication Systems
- Digital Signal Processing
- EV and Renewable Energy

PROFESSIONAL EXPERIENCE

Samsung Semiconductor, Bangalore (May'20-June'20)	Project: Positive Charge Pump and LDO Circuit Design -Designed cascaded voltage doubler for positive charge pump and studied its disadvantages. -Designed cross coupled positive charge pump for improved efficiency. -Designed array of cross coupled charge pumps to support higher load currents. -Designed non overlap clock generator circuit. -Designed model LDO circuit for regulation of charge pump output.
Indian Institute of Science (IISc), Bangalore (May'19-July'19)	Project: Analog Neuromorphic Circuit Design -Synthesized bias scalable CMOS analog computational circuits using Margin Propagation. -Designed a perceptron as a linear pattern classifier using an inner product approximate circuit. -Designed a Multilayer Perceptron using Margin Propagation operating in subthreshold region for classification of non linearly separable data using ReLU as activation function.
ISRO, SDSC, Sriharikota (May'18-June'18)	Project: Matched Filter Bank in RADAR Signal Processing -Implemented a bank of matched filters in Matlab for use in processing return signals (echoes) received from a remote target in space. -Created an efficient interpolation technique that generates an estimate of the Doppler frequency of the target based on output signals of the bank of matched filters

PROJECTS

Phase Locked Loop (EE6324)	-Designed single ended ring oscillator using inverters. -Designed pseudo differential ring oscillator.
Digital IC Design (EE5311)	-Designed layout and schematic of 8 bit Signed Carry Save Multiplier (CSM) in Electric with ripple adder as the vector merge stage. -Improved propagation delay of the circuit using Carry Lookahead Adder .
Analog IC Design (EE5320)	-Designed single stage 5-T differential amplifier circuit. -Designed 2-stage differential Op-Amp circuit with millar compensation . -Designed common source amplifier (CSA) circuit.
Analog Systems Lab (EE2019)	-Built a composite system to detect synchronised light and sound and selectively amplified 1kHz and 3kHz light and sound signals. -Designed LED driver based on DC-DC converter, peak detector and bandpass filters.

POSITIONS OF RESPONSIBILITY

Deputy Coordinator, Placement Team	-Coordinated with students and recruiters to organize and manage campus placements for the year 2017. -Convinced new companies to take part in the recruitment process of IITM. -Organised tests, presentations and interviews before and during Placement Season, 2017.
Coordinator, Branding and Engagement, CFI	-Organised the CFI Open House, 2017 and the CFI Open House launch program. -Organised the event that set the record for 'Largest number of Robots cleaning an area' appearing in Asia Book of Records and India Book of Records.
Coordinator, Inter IIT Sports Meet	-Part of the organising team at Inter IIT Sports Meet held at IIT Madras in 2017. -In charge of room allocation and hospitality arrangement for players at the meet.

EXTRA- CURRICULAR ACTIVITIES

Dramatics, Short Film Making, Elocution	-Participated in various full length play and skits and organised Inter College Dramatics Competition, Saarang, 2016. -Secured third position in Short Film Competition, LitSoc, 2017. -Secured fourth position in Elocution Competition, LitSoc, 2017.
Sports	-Member of Sharavati Hostel Table Tennis Team, Schroeter.